

Protective effect of sulforaphane against oxidative stress: recent advances.

Sulforaphane [1-isothiocyanate-(4R)-(methylsulfinyl)butane] is a natural dietary isothiocyanate produced by the enzymatic action of the myrosinase on glucopharanin, a 4-methylsulfinylbutyl glucosinolate contained in cruciferous vegetables of the genus Brassica such as broccoli, brussel sprouts, and cabbage. Studies on this compound is increasing because its anticarcinogenic and cytoprotective properties in several in vivo experimental paradigms associated with oxidative stress such as focal cerebral ischemia, brain inflammation, intracerebral hemorrhage, ischemia and reperfusion induced acute renal failure, cisplatin induced-nephrotoxicity, streptozotocin-induced diabetes, carbon tetrachloride-induced hepatotoxicity and cardiac ischemia and reperfusion. This protective effect also has been observed in in vitro studies in different cell lines such as human neuroblastoma SH-SY5Y, renal epithelial proximal tubule LLC-PK1 cells and aortic smooth muscle A10 cells. Sulforaphane is considered an indirect antioxidant; this compound is able to induce many cytoprotective proteins, including antioxidant enzymes, through the Nrf2-antioxidant response element pathway. Heme oxygenase-1, NAD(P)H: quinone oxidoreductase, glutathione-S-transferase, gamma-glutamyl cysteine ligase, and glutathione reductase are among the cytoprotective proteins induced by sulforaphane. In conclusion, sulforaphane is a promising antioxidant agent that is effective to attenuate oxidative stress and tissue/cell damage in different in vivo and in vitro experimental paradigms. Copyright © 2010 Elsevier GmbH. All rights reserved.